

western University, in 1996.

Cobots are an improved version of robots. These deliver more accuracy to work, speed up tasks, and bridge the gap between manufacturers and customers. Current robotic systems are expensive and need more skills to operate. Whereas, training cobots is easy because humans can interact with these without using any programming language or script. This results in ill-equipped manufacturing systems being able to handle errors better.

AR allows viewing of information that otherwise resides on screens or paper. It makes processes faster and more accurate. AR can be used with digital twin technology with a device model. It sets about exploring ways to improve information delivery to users.

Combining AR and cobots can lead to some interesting applications, further enhancing robot and human interaction. However, such systems are still in their research phase.

Popular examples of AR are WorkLink and Google Glass. Rittal is using AR at its research centre to assist tasks such as wiring, debugging and process control. The company is integrating AR with its EPLAN platform builder software. EPLAN is used to design and model almost everything, from wiring harnesses to control panels. An AR system can significantly reduce errors that may occur due to incorrect wiring, by showing user information with visual context. It can speed up wiring processes as well.

AR-based solutions are also being used for training. These reduce training times, because information is readily available. AR solutions depend on enterprise-class reality platforms that use speech recognition and voice biometrics to enable automatic information delivery and capture.

Cobots come in many forms. Modular individual transport (MIT) cobot can lift up to 500 kilograms. It comes with multiple drive and lift options. Multiple cobots can be synchronised for coordination of movement. The system can be equipped with sensors to avoid obstacles and aid navigation. MIT cobot can be controlled remotely or integrated with an autonomous guidance system.

Both AR and cobots are changing the face of industrial operations.

Apple and Google released AR software development kits (SDKs) for their respective platforms. Many mixed-reality headsets support this software framework, including devices from HP, Dell, Acer and Samsung.

In gaming arena, AR glasses provide an immersive video environment. Many AR devices that were showcased at Consumer Electronics Show 2018 can be utilised as a part of solutions, from simulating automotive experiences to control drones.

Then, there is Snappy, an AR app for smartphones, using which users can record geo-specific messages complete with animated AR characters.

Cobots can be used in controlled spaces like hospitals. Aethon's TUG cobot has been walking around medical facilities for years. TUG's lidar and other sensors are similar to Marble's. The former uses Wi-Fi to communicate with doors and elevators. Aethon's family of TUG cobots have found a home in controlled environments like hospitals, delivering everything from medicine to food.

Sophia, the famous humanoid cobot, has the world amazed by her gentle nature and compelling intelligence. Developed by Hong-Kong based Hanson Robotics, Sophia is quite a marvel to watch. Saudi Arabia even offered her citizenship.

Rethink Robotics' Sawyer is an armed cobot designed to operate in close proximity with humans. Like many cobots, Sawyer detects humans through sight, sensors and touch. It also has limbs that maximise its powers. Sawyer comes with a counter weighted arm for low-impact operations.

Easy movement of arms by a person is one way of programming cobots like Sawyer and Baxter. It enables their training programs to move the arm around, adjust grippers, manipulate objects for the camera and so on.

Feston's Bionic cobot is a pneumatic robot that has seven joints.

Types of cobots

There are four types of cobots, as defined by ISO/TS 15066 standard.

These are power and force-limiting, safety-monitoring, hand-guiding, and speed and separation monitoring, with most people focusing on just the first.

The main difference between a power and force-limiting cobot and the other three is that the former can use regular industrial robots with additional sensors and still be totally safe for human-robot collaboration. The power and force-limiting cobot represents robots that can sense forces and stop to avoid injuries. It has all kinds of sensors embedded in it.

Force-limiting cobots are of following types:

Joint-sensing. It is the most common type of cobot that includes robots like Universal Robot (UR 3, UR 5 and UR 10) and KUKA (IIWA). It uses joints to monitor the forces applied to its body.

Force-sensor base. This cobot has a different way of feeling the forces applied to its body. As the name suggests, large force-torque sensors at its base monitor the different forces. However, payload needs to be set correctly to allow the robot to compute whether the force applied at the force sensor is higher than the weight it is supposed to handle. This means that a force-sensor base robot is more complicated to use than a joint-sensing one. But, it is safer and more precise.

Skin-sensing. This force-limiting robot is not as widespread in the industry, but is probably the safest option out there. Robots like Bosch (APAS) use tactile sensing (of all kinds) to detect impact. Sensors monitor the conductivity of the robot's body and instructs it to stop once it reaches a certain threshold.

Inherently-safe. An inherently-safe robot may use any (or all) sensor(s). What distinguishes it from the others is that it cannot hurt users in any way.

Cobots and AR in India

The first cobot arrived in India in 2011, when Bajaj Auto deployed it in one of its factories.

Aurolab is a Madurai-based company that makes intraocular lenses.